
chapter four

PROBABILITY

*This branch of mathematics [probability] is the
only one, I believe,
in which good writers frequently get results
entirely erroneous.*

– Charles Sanders Pierce

INTRODUCTION

This chapter on probability is often the most challenging chapter in the introductory statistics course. In many respects, one could do a minimal amount of work in probability and still be able to do most of the problems in statistics. This, of course, brings us to the question of “Why do we have this chapter on probability?”

In part, the answer to the above question involves the creation and development of the introductory statistics course in the 1960’s. At that time, the usual statistics curriculum was taught in upper division and involved a semester of probability; followed by a second semester of statistics. When the lower division statistics course evolved, a chapter on probability was considered to be an important part of the course.

Aside from the historical perspective, there is good reason to understand the ideas in the probability chapter. Our early work in constructing relative frequency histograms and polygons from the relative frequencies of grouped data provided us with what are called experimental or *a posteriori* probabilities. It only seems natural to extend

this idea in a formal way to theoretical or *a priori* probabilities.

Theoretical probabilities are developed using the idea of a **sample space** that results from the outcomes of the experiment. We will focus on the outcomes, called **simple events**. Capital letters, such as A and B are used to represent **events** that are collections or sets of the simple outcomes.

Events are classified in a variety of ways using terms that have an intuitive appeal that helps us understand their meaning. For example, we will talk about events being **independent**, **mutually exclusive**, and **complementary**. We then find the probability of events occurring using the idea of counting. Unfortunately, the counting process can become cumbersome. It is simplified through the use of **Tree Diagrams**, **Venn Diagrams**, the **Counting Rule** and formulas that focus on the probability of two events occurring. Specifically, we will be interested in the probability of the **compound events** (A and B) and (A or B), and the **conditional probability** of A given B .

We then apply the concepts and formulas by doing problems that involve experiments such as tossing coins, drawing cards, and throwing dice. The problems provide us an opportunity to become familiar with the terms, formulas, and notation used in probability.

Many of the ideas we develop in this chapter are essential to understanding the work we do in the chapters that follow. For example, in the next two chapters, we use the concept of probability in defining both discrete and continuous probability distributions. These theoretical distributions are used as models for the data distributions we worked with earlier.

We use the distributions in the hypothesis testing that is such an important part of inferential statistics. In fact, the very essence of hypothesis testing involves finding the probability of getting the value of the sample statistic we use to challenge the accepted value of the population parameter.

CHAPTER 4 HOT SPOTS

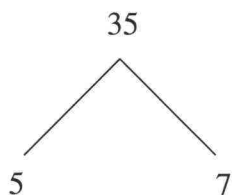
1. **Using Tree Diagrams.**
Starts on page 4–5. Problems on pages 4–10, 4–11, 4–12.
Mann 4.1
2. **Using The Counting Rule.**
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3. **Using The Complement.**
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4. **Independent Events Versus Dependent Events.**
Starts on page 4–17. Problems on pages 4–19, 4–20. Mann 4.6, 4.8.2
5. **Mutually Exclusive Versus Independent Events.**
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6. **Special Vs. General Formula For $P(A \text{ and } B)$ & $P(A \text{ or } B)$.**
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7. **Deciding On Which Formula To Use.**
Starts on page 4–22. Problems on pages 4–23, 4–24.
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8. **$P(A \text{ and } B)$ Versus $P(A|B)$.**
Starts on page 4–25. Mann 4.8.2

If you find other HOT SPOTS, write them down and use them as a focus of your discussions in the study group. Or you can use the HOT SPOT as the topic for a help session with your professor.

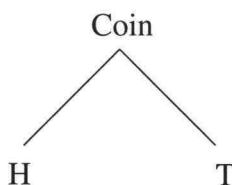
USE THIS PAGE FOR KEEPING TRACK OF TERMS AND NOTATION

Hot Spot #1 – Using Tree Diagrams

We will start with the Tree Diagram. It is perhaps the most important topic in this chapter. The Tree Diagram is very similar to a factor tree that we first see in the prime factoring of composite numbers. For example, in factoring 35 we get the following factoring tree:

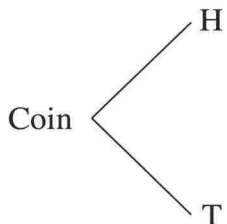


The factor tree branches out to the two possible factors. We consider the toss of one coin to be like starting with a number. The two outcomes of H (heads) and T (tails) are like the factors. We diagram the outcomes just as we did the factoring:



We usually turn the Tree Diagram on its side as illustrated below:

Tree Diagram



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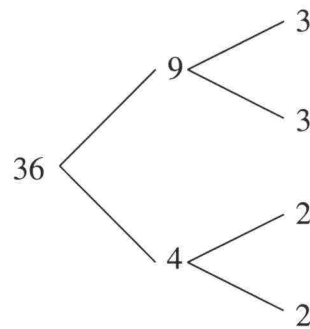
Figure 4-1:
 5×7 Factor
Tree

Figure 4-2:
Coin Toss
Diagram

Figure 4-3:
Coin Toss Tree
Diagram

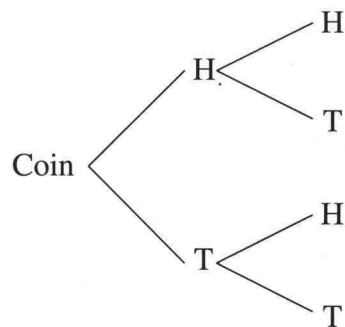
We can treat the toss of two coins like the factoring of 36. The second factoring resembles the second toss of the coin. The list of factors in the last step is like the list of outcomes we get in the tree diagram. The two diagrams are illustrated below:

Factor Tree



*Figure 4-4:
Factor Tree*

Tree Diagram



*Figure 4-5:
Tree Diagram*

We now complete the Tree Diagram by listing the probability of each outcome on the appropriate branch:

Tree Diagram

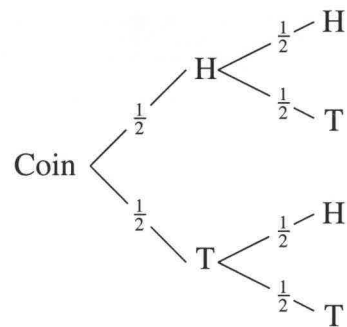


Figure 4-6:
Tree Diagram
w/Probabilities

The Tree Diagram can be used to help us list both the outcomes of an experiment and the probability of each of the outcomes:

Tree Diagram

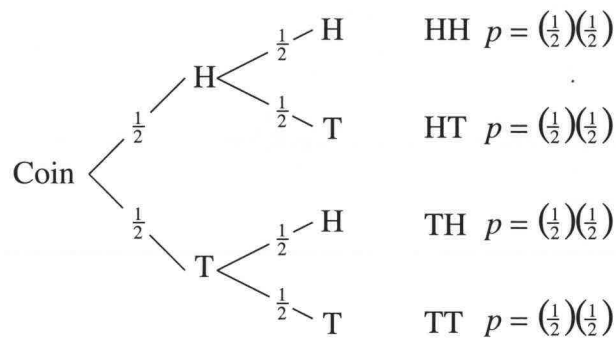


Figure 4-7:
Tree Diagram
w/Outcomes

There are times when all we do is list outcomes without writing in the probabilities of each outcome. In fact, we may not know the probabilities. We then simply use the Tree Diagram as a tool to help us understand how to do a problem. The following Tree Diagram, without the probabilities written in, illustrates the toss of one die (the singular of "dice"):

Tree Diagram

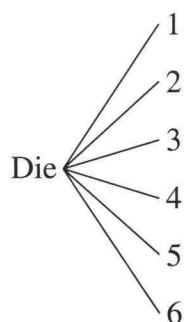


Figure 4-8:
Tree Diagram,
Toss Of One Die

If we were to toss the die a second time the Tree Diagram would be more difficult to draw and require more space. It is drawn on the following page as a solution to the first **Sample Problem and Solution**. The solution will take a full page to draw. This time, neither the probabilities nor the outcomes of the experiment will be listed.

The Tree Diagram is perhaps most useful when we are illustrating probability experiments that involve drawing without replacement. When we include the probabilities for each branch and the list of outcomes for the experiment, the diagram provides us the answers for questions about conditional probability.

As an example, we will draw two cards without replacement from a regulation deck containing 52 cards with four suits and four aces. We are interested in drawing an ace. Each time we draw, we will either get an ace (A) or not an ace ($\bar{A}$). The probability of drawing A on the first draw is $\frac{4}{52}$. But the probability of A on the second draw depends on what happened on the first draw. The probability of $\bar{A}$ on the first draw is $\frac{48}{52}$. But again, the probability of $\bar{A}$ on the second draw depends on what happened on the first draw. The following Tree Diagram helps us organize what is starting to sound complicated:

Tree Diagram

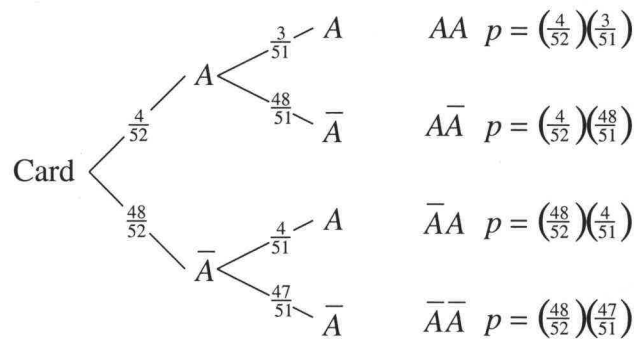


Figure 4-9:
Tree Diagram,
Draw of Two
Cards

Hot Spot #1 Sample Problem and Solution: Draw the Tree Diagram for two tosses of a six sided die.

Solution:

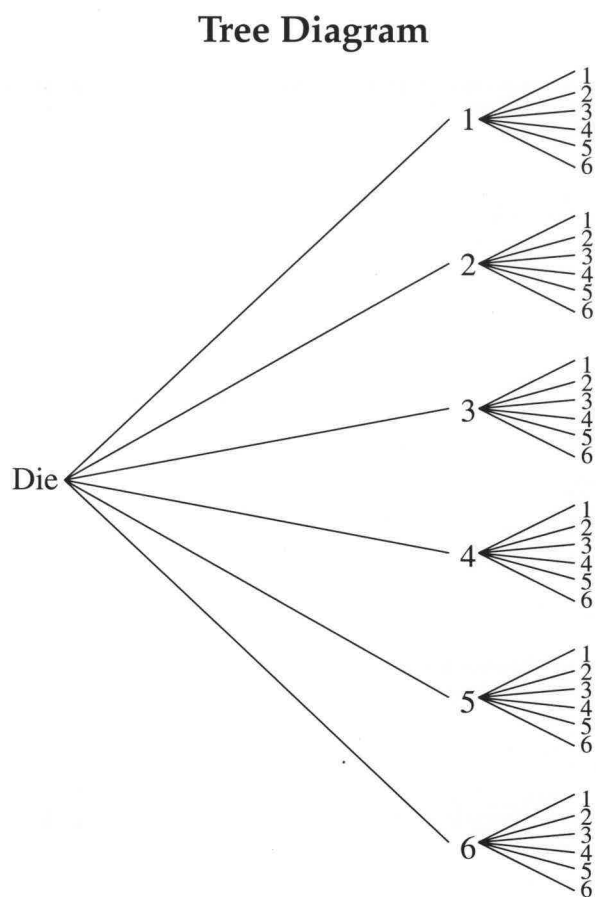


Figure 4-10:
Tree Diagram,
Two Tosses Of A
Die

Sample Problem and Answer: Draw a Tree Diagram for three tosses of a coin. Include the probabilities and the outcomes of the experiment.

Answer:

Tree Diagram

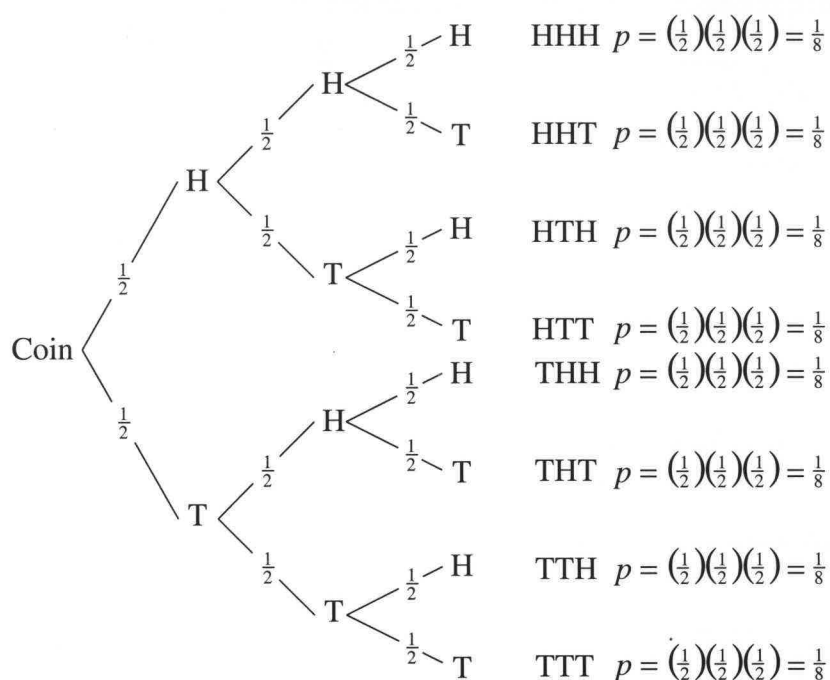


Figure 4-11:
Tree Diagram,
Three Tosses Of
A Coin

Sample Problem and Answer: Draw the Tree Diagram for the experiment consisting of tossing one coin followed by the draw of one card from a deck where success is drawing an ace. Include the individual probabilities for each branch, the outcomes of the experiment, and their probabilities in the Tree Diagram.

Answer:

Tree Diagram

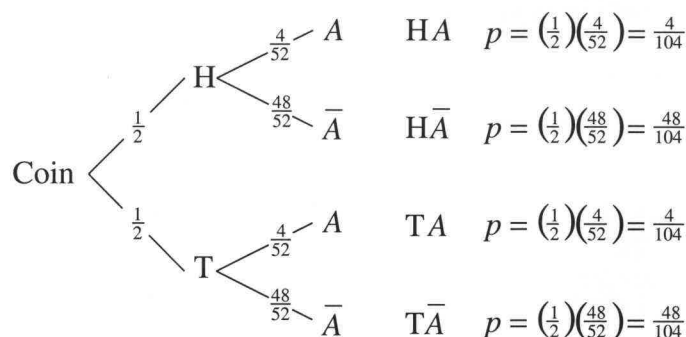


Figure 4-12:
Tree Diagram,
Toss Of One
Coin And Draw
Of One Card



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Hot Spot #2 – Using The Counting Rule

On the surface, probability seems straight forward. After all, we only need to list the sample space and count which outcomes are a success. Unfortunately, listing outcomes is not always easy; even with a Tree Diagram. For example, the popular dice game, YAHTZEE, involves tossing five dice. This is the same as taking one die and tossing it five times. When we get all five dice with the same number we have a YAHTZEE! It is interesting to find the probability of a YAHTZEE, because it is unreasonable to list the outcomes using a Tree Diagram. Yet, it is easy to find the answer.

What we do is find a way of determining the number of outcomes without counting them. We know that there are only six ways of getting a YAHTZEE. All we need is the number of total possible outcomes to write the correct probability.

In the first **Sample Problem and Solution, (Hot Spot #1)** we found that there were 6×6 , or 36, outcomes when we used two dice. All we have to do is realize that for every new dice we toss we have six new outcomes. The six new outcomes will be multiplied by the previous six outcomes. When we toss five dice we will have $6 \times 6 \times 6 \times 6 \times 6$, or 7,776 total possible outcomes. The probability of a YAHTZEE is $\frac{6}{7,776}$.

$$P(\text{YAHTZEE}) = \frac{6}{7,776}$$

The principle we just used is called the counting rule. Let us assume we do two things at once where there are k ways of doing the first thing and m ways of doing the second thing. There are then km ways of doing the two things at the same time. If we extend to n ways for doing a third thing, then we would have kmn ways of doing the three things together. The counting rule gives us a way of often finding the number of total outcomes without actually counting or listing the outcomes with a Tree Diagram.

*The Counting
Rule*

Hot Spot #2 Sample Problem and Solution: How many outcomes are there when we toss 5 coins?

Solution: There are two outcomes for each coin. Since we are tossing five coins, we multiply $2 \times 2 \times 2 \times 2 \times 2$ to get 32 outcomes.

Sample Problem and Answer: We toss a six sided die and draw a card from a regulation deck of 52 cards where we record the suit drawn. How many outcomes are possible?

Answer: $6 \times 4 = 24$ outcomes.

Sample Problem and Answer: An experiment involves tossing a coin, tossing a six sided die, and drawing a card from a deck of 52 cards and recording the suit. How many possible outcomes?

Answer: $2 \times 6 \times 4 = 48$ outcomes.

Hot Spot #3 – Using The Complement

The YAHTZEE problem we did earlier in **Hot Spot #2** provides us with another insight regarding complementary events. The complement of an event A is everything in the sample space after A is removed. Since there were only six ways of getting a YAHTZEE, there are $7,776 - 6 = 7,770$ outcomes in the complement of the set of six outcomes that gave us a YAHTZEE. With this information, we can find the probability of not getting a YAHTZEE.

$$P(\text{Not a YAHTZEE}) = \frac{7,770}{7,776}$$

A simpler way of finding the probability of “not a YAHTZEE” is to use the probability formula for complementary events:

$$P(\bar{A}) = 1 - P(A)$$

*Complementary
Events Formula*

Using the above formula,

$$P(\bar{Y}) = 1 - P(Y) = 1 - \frac{6}{7,776} = \frac{7,770}{7,776}$$

The probability formula for complementary events is very useful when the problem we are trying to do has many outcomes. The complement of the event then involves few outcomes. Finding the probability of at least one success is one of the classic problems using the idea of a complement. For example, suppose we are tossing a coin three times and we want the probability of at least one tail. We use the complement of at least one tail which is getting zero tails.

$$P(\text{at least one tail}) = 1 - P(\text{zero tails})$$

There is only one way of getting zero tails, namely getting exactly three heads. The answer is easy to set up as follows:

$$P(\text{at least one tail}) = 1 - \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = 1 - \frac{1}{8} = \frac{7}{8} = 0.875$$

The key phrase that most often helps us know when to use the complement is “find the probability of at least one success.”

Hot Spot #3 Sample Problem and Solution: In a toss of a die three times, what is the probability of getting at least one 3?

Solution: The complement of getting at least one 3 is to get zero 3's. This can be done in only one way. The probability of not getting a 3 is $\frac{5}{6} = 0.83\bar{3}$.

$$\begin{aligned}P(\text{at least one } 3) &= 1 - P(\text{zero } 3\text{'s}) \\&= 1 - \left(\frac{5}{6}\right)\left(\frac{5}{6}\right)\left(\frac{5}{6}\right) = 1 - 0.58 = 0.42\end{aligned}$$

Sample Problem and Answer: In a production process ten items are selected from a shipment. Assume that each item has a probability of 0.05 of being defective. What is the probability that at least one item will be defective?

Answer: We note that $P(\text{zero defectives}) = 0.95$ for each selection.

$$\begin{aligned}P(\text{at least one defective}) &= 1 - P(\text{zero defectives}) = 1 - (0.95)^{10} \\&= 1 - 0.599 = 0.401\end{aligned}$$

Sample Problem and Answer: In five tosses of a coin what is the probability of at least two heads?

Answer: The probability of at least two heads tells us we are interested in $x \geq 2$, where x is the number of heads. The complement of $x \geq 2$ is $x \leq 1$. Thus the complement of at least two heads is the event of getting either zero heads or one head. The

$P(\text{zero heads}) = (\frac{1}{2})^5$ and the $P(\text{one head}) = 5(\frac{1}{2})^5$. If we looked at a Tree Diagram for five tosses of a coin, we would find that five of the thirty-two branches have exactly one head.

$$\begin{aligned} P(\text{at least two heads}) &= 1 - (P(\text{zero heads}) + P(\text{one head})) \\ &= 1 - \left((\frac{1}{2})^5 + 5(\frac{1}{2})^5 \right) = 1 - \left(\frac{1}{32} + \frac{5}{32} \right) \\ &= 1 - \frac{6}{32} = 1 - 0.1875 = 0.8125 \end{aligned}$$

Hot Spot #4 – Independent Events Versus Dependent Events

The work with conditional probability leads us to the idea of independent events versus dependent events. The idea of independence suggests that one event does not influence a second event. However, the concept is really defined using a formula irrespective of whether or not the independence appeals to us intuitively. The definition for independent events A and B requires the following to be true:

$$P(A|B) = P(A) \text{ and } P(B|A) = P(B)$$

The two statements are equivalent. For example, If we know $P(A|B) = P(A)$, then we know $P(B|A) = P(B)$. If $P(A|B) \neq P(A)$ or $P(B|A) \neq P(B)$, the events are dependent.

We can illustrate the idea with the following example that involves tossing a six sided die. Find the probability that we get a 3, given we know the outcome is a multiple of 3.

$$\text{Let } A = \{3\} \quad B = \{3, 6\} \quad S = \{1, 2, 3, 4, 5, 6\}$$

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Sections
4.6 And 4.8.2

Independent
Events

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{\frac{1}{6}}{\frac{2}{6}} = \frac{1}{2} \qquad P(A) = \frac{1}{6}$$

$$P(A|B) \neq P(A)$$

So A and B **are not** independent events; they are dependent events.

In contrast, we will consider a similar problem again using a six sided die. This time we want the probability of an even number, given we know the outcome is a multiple of three.

$$\text{Let } A = \{2, 4, 6\} \quad B = \{3, 6\} \quad S = \{1, 2, 3, 4, 5, 6\}$$

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{\frac{1}{6}}{\frac{2}{6}} = \frac{1}{2} \qquad P(A) = \frac{1}{2}$$

$$P(A|B) = P(A)$$

So A and B **are** independent events.

We need to note that the last two problems were alike in structure, and the given information was the same. In the first problem, the events were dependent, but in the second problem the events were independent. The point is we must check for independence using the following definition:

$$P(A|B) = P(A) \text{ if and only if } A \text{ and } B \text{ are independent.}$$

*Independent
Events Formula*

Hot Spot #4 – Sample Problem and Solution: Consider the toss of two six sided dice. Let A equal the event *the sum is even* and B equal the event *the sum is a multiple of three*. Determine if A and B are independent.

Solution: There are 36 pairs of outcomes when throwing two dice. (See the tree diagram on page 4–12 of the Study Guide).

$$P(A) = \frac{18}{36} \quad P(A \text{ and } B) = \frac{6}{36} \quad P(B) = \frac{12}{36} \quad P(A|B) = \frac{6}{12}$$

Since $P(A) = \frac{1}{2}$ and $P(A|B) = \frac{1}{2}$, A and B are independent.

Sample Problem and Answer: Given the toss of one six sided die, let A equal the event *the number is more than three* and let B equal the event *the number is a multiple of three*. Determine if A and B are independent.

Answer: $P(A) = \frac{3}{6} = \frac{1}{2}$ and $P(A|B) = \frac{1}{2}$. Therefore, A and B are independent.

Sample Problem and Answer: Consider the toss of a six sided die. Let A equal the event *the number is even* and B equal the event *the number is odd*. Check for independence.

Answer: $P(A) = \frac{3}{6}$ $P(A|B) = \frac{0}{6}$. Therefore, A and B are dependent.



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Sections
4.5, 4.6, And
4.8.2

Mutually
Exclusive
Events

Hot Spot #5 – Mutually Exclusive Versus Independent Events

When A and B cannot both occur together, we say the two events are mutually exclusive. Another way of looking at the idea is that the first event excludes the second event. This is illustrated in tossing a die and letting A = odd number and B = even number with $S = \{1, 2, 3, 4, 5, 6\}$. Events A and B are said to be mutually exclusive. Whenever we have $P(A|B) = 0$ the two events are said to be mutually exclusive:

$$P(A|B) = 0 \text{ because } P(A \text{ and } B) = \frac{n(A \text{ and } B)}{n(S)} \text{ and } n(A \text{ and } B) = 0.$$

It is important to understand the difference between events that are mutually exclusive and those that are independent. A simple way of relating the two ideas is that if two events are mutually exclusive they must be dependent; and therefore not independent. For example, consider two events A and B that are mutually exclusive, with $P(A) = 0.5$:

$$P(A \text{ and } B) = 0 \text{ because } A \text{ and } B \text{ are mutually exclusive.}$$

Using the formula for $P(A|B)$, we get

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{0}{P(B)} = 0.$$

Since $P(A) = 0.5$ and $P(A|B) = 0$, the events are dependent.

We must be careful not to get careless in our thinking. It may seem that mutually exclusive events and dependent events are one and the same. This is clearly **not** the case. The following example involves two events that are dependent but not mutually exclusive. The problem of finding the probability of getting a 3 given a multiple of 3 on the toss of one die is done as follows:

Let $A = \{3\}$ $B = \{3, 6\}$ with $S = \{1, 2, 3, 4, 5, 6\}$

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{\frac{1}{6}}{\frac{2}{6}} = \frac{1}{2} \quad P(A) = \frac{1}{6}$$

Since $P(A|B) \neq P(A)$, A and B are dependent.

Since $P(A \text{ and } B) \neq 0$, A and B are **not** mutually exclusive.

Hot Spot #6 – Special Vs. General Formula For $P(A \text{ and } B)$ & $P(A \text{ or } B)$

There are two formulas in the chapter that we have not used so far in this review. They are referenced as the addition and the multiplication rules. Actually, we have used the formulas informally as we have worked through some of the problems.

When we considered the probability of events A or B , we simply added the probabilities when the events were mutually exclusive. But not all events are mutually exclusive; so we need what is called the addition rule that is used when we want the probability of A or B .

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

In a similar way, when we worked with Tree Diagrams and considered the probability of A and B , we multiplied the probabilities together. If we do not use a Tree Diagram, it may not be clear whether one event is influencing another; as it happened when we drew cards without replacement. As a result, we need what is called the multiplication formula that is used when we want the joint probability of A and B .

$$P(A \text{ and } B) = P(A)P(B|A)$$

or equivalently

$$P(A \text{ and } B) = P(B)P(A|B)$$

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4.8.2 And 4.9.2

$P(A \text{ or } B)$
General Case

$P(A \text{ and } B)$
General Case

$P(A \text{ or } B)$
Special Case

In the special cases of independent events and mutually exclusive events, the addition and multiplication formulas can be written in much simpler forms.

If events A and B are mutually exclusive, the joint probability $P(A \text{ and } B) = 0$. The addition rule then becomes the sum of the marginal probabilities:

$$P(A \text{ or } B) = P(A) + P(B)$$

$P(A \text{ and } B)$
Special Case

If events A and B are independent, $P(B|A) = P(B)$, then the multiplication rule for the joint probability of A and B becomes the product of the marginal probabilities:

$$P(A \text{ and } B) = P(A)P(B)$$

Clearly, it is an advantage to be aware of the special cases. But as a rule, it is better to learn the more general formulas that will work for any events, whether or not they are mutually exclusive or independent.

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4.4, 4.5, 4.6, 4.7,
4.8, and 4.9

Hot Spot #7 – Deciding On Which Formula To Use

One of the most difficult things to do in this chapter is to decide on which formula to use. There are a few keys that will help us decide on the correct formula. For example, the use of the phrase “probability of at least one” suggests the complement formula $P(\bar{A}) = 1 - P(A)$. The phrase “determine if events are independent” tells us to use

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}.$$

Another way to help us decide on the correct formula is to become familiar with the formulas. A good way to do this is to do problems that require we know and correctly use appropriate formulas. The following example illustrates the idea:

Given $P(A) = 0.5$, $P(A \text{ and } B) = 0.25$; find $P(B|A)$.

Step 1. We need a formula that contains $P(B|A)$ so we use $P(A \text{ and } B) = P(A)P(B|A)$.

Step 2. We substitute the given information into the formula, getting $0.25 = (0.5)P(B|A)$.

Step 3. We then solve for $P(B|A) = \frac{0.25}{0.5} = 0.5$

Hot Spot #7 Sample Problem and Solution: Given $P(A \text{ and } B) = 0.25$, $P(A) = 0.5$, $P(A \text{ or } B) = 0.7$; find $P(B)$.

Solution: Step 1. We need a formula that includes $P(A)$, $P(A \text{ and } B)$, $P(A \text{ or } B)$ and $P(B)$, so we use

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B).$$

Step 2. We substitute getting $0.7 = 0.5 + P(B) - 0.25$.

Step 3. We then solve for $P(B) = 0.7 - 0.5 + 0.25 = 0.45$.

Sample Problem and Answer: Given $P(\bar{A}) = 0.3$; find $P(A)$.

Answer: Step 1. We need the formula for the complement

$$P(A) = 1 - P(\bar{A}).$$

Step 2. We substitute, getting $P(A) = 1 - 0.3 = 0.7$.

Sample Problem and Answer: Given $P(A|B) = 0.2$, $P(\bar{B}) = 0.3$, find $P(A \text{ and } B)$.

Answer: Step 1. We need two formulas: $P(B) = 1 - P(\bar{B})$ and

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}.$$

Step 2. We use the first formula and solve for $P(B)$.

$$P(B) = 1 - 0.3 = 0.7$$

Step 3. We then use the second formula and substitute the value for $P(B)$ and $P(A|B)$.

$$0.2 = \frac{P(A \text{ and } B)}{0.7}$$

Step 4. We solve.

$$P(A \text{ and } B) = (0.2)(0.7) = 0.14$$

Hot Spot #8 – $P(A \text{ and } B)$ Versus $P(A|B)$

There is one particularly confusing situation involving the multiplication formula. The formula can be written in two ways as follows:

$$P(A \text{ and } B) = P(A)P(B|A) \text{ or } P(B|A) = \frac{P(A \text{ and } B)}{P(A)}.$$

In the first form we are looking at the joint probability of A and B , and we call it the multiplication rule. But in the second form, we are focusing on the probability of A given B . We get the second form by using the first formula and solving for $P(B|A)$. The second form is referred to as the formula for conditional probability.

The confusion occurs when we are asked to find $P(B|A)$, and we realize we need $P(A \text{ and } B)$. It seems like we should be able to use the first formula to find $P(A \text{ and } B)$. But we then realize that this requires $P(B|A)$, which is what we wanted in the beginning. We have just gone full-circle; like a puppy chasing its tail. We are not able to do what we wanted to do. Instead, we must go back to the sample space to find the $P(A \text{ and } B)$. We then substitute our value into the equation

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)}$$

to get our answer.

Another problem occurs when we realize that there are really two variations of the formula for what we call conditional probability. We can write either

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)} \text{ or } P(B|A) = \frac{P(A \text{ and } B)}{P(A)}.$$



Mann
Section 4.8.2

Conditional
Probability
Formula

The situation is not so nearly confusing if we note the following:

- In the phrase $P(A|B)$, the $A|B$ reminds us of the fraction

$$\frac{A}{B},$$

whose denominator is B .

- On the right side of the equal sign we have

$$\frac{P(A \text{ and } B)}{P(B)},$$

whose denominator involves the same letter B .

We can restate the pattern as follows:

The letter in the denominator on the right side of the equal sign is the same as the letter that follows the vertical bar () on the left side of the formula.

*Using A Pattern
To Find The
Denominator In
The Formula For
Conditional
Probability*

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)}$$

↑ letter after vertical bar

← letter in denominator on right side

CHAPTER 4 DISCUSSION QUESTIONS

These questions may be used in your study group or simply as topics for individual reflection. Whichever you do, take time to explain verbally each topic to insure your own understanding. Since these questions are intended as topics for discussion, answers to these questions are not provided. If you find that you are not comfortable with either your answers or that your group has difficulty with the topic, take time to meet with your professor to get help.

1. What is the difference between dependent and independent events?
2. Why is the probability of A and B equal to zero when A and B are mutually exclusive?
3. When using a Tree Diagram, why is the number of branches at the end the same as the number of total possible outcomes?
4. What is the difference between marginal probability and conditional probability?
5. What is the difference between the relative frequency concept and the classic definition of probability?
6. When does one use the probability formula $P(A) = 1 - P(\bar{A})$?
7. When is the formula $P(A \text{ or } B) = P(A) + P(B)$ used?
8. When is the formula $P(A \text{ and } B) = P(A)P(B)$ used?
9. When drawing from a container without replacement, what form of the formula for the probability of A and B is used?
10. What rule is used to help decide the number of possible outcomes for a probability experiment?